UNIT 1 · TOPIC 2(A)

Power-Operated Equipment — Draft, Pull and Power

Syllabus topic: Functional requirement, principle of working and constructional features of animal and power operated equipment

These working definitions apply to every power-operated implement in Unit 1.

Key definitions

Term	Meaning
Pull	Total force the tractor exerts on the implement, along the line of hitch.
Draft	Horizontal component of pull, in the direction of travel.
Unit (specific) draft	Draft per unit cross-sectional area of the furrow (kgf/cm² or N/cm²).
Drawbar power	Draft × forward speed.

Formulas

$$\text{Draft} = \text{Pull} \times \cos\theta$$

$$\text{Unit draft} = \text{Draft} \div \text{furrow area } (W \times d \text{ per bottom} \times n)$$

$$\text{Drawbar power (kW)} = \text{Draft (kN)} \times \text{Speed (m/s)}$$

$$\text{DBP (hp)} = (\text{Draft in kgf} \times \text{Speed in m/s}) \div 75$$

Working it out, step by step

1. Measure the pull along the line of hitch and the hitch angle θ.
2. Find the draft: Draft = Pull × cos θ.
3. For unit draft, divide the draft by the furrow cross-sectional area.
4. For drawbar power, multiply the draft by the forward speed.

How draft changes with speed (Kepner)

$$D = a + bV^2$$

Symbol	Meaning
D (Draft)	Total horizontal force in the direction of travel needed to pull the implement (kN or lbf).
V (Velocity/Speed)	Forward operating speed of the tractor and tool (km/h or mph).
a (Static component)	Constant for the baseline soil resistance at near-zero speed. It depends heavily on implement type, soil texture, compaction and depth of cut.
b (Dynamic component)	Multiplier on the speed-squared term. It accounts for the dynamic forces needed to accelerate the sheared soil.

- Typical unit draft: light sandy soils **~0.2–0.4 kgf/cm²**; heavy clays **0.7–1.4 kgf/cm²**.
- Typical speeds: primary tillage **4–6 km/h**; secondary tillage **5–8 km/h**.

Practice MCQs

Q22. Draft of an implement equals:

(a) Pull × sin θ
(b) Pull × cos θ
(c) Pull × tan θ
(d) Pull ÷ cos θ

Answers

Q22 — (b) Pull × cos θ — *Draft is the horizontal, direction-of-travel component of pull.*